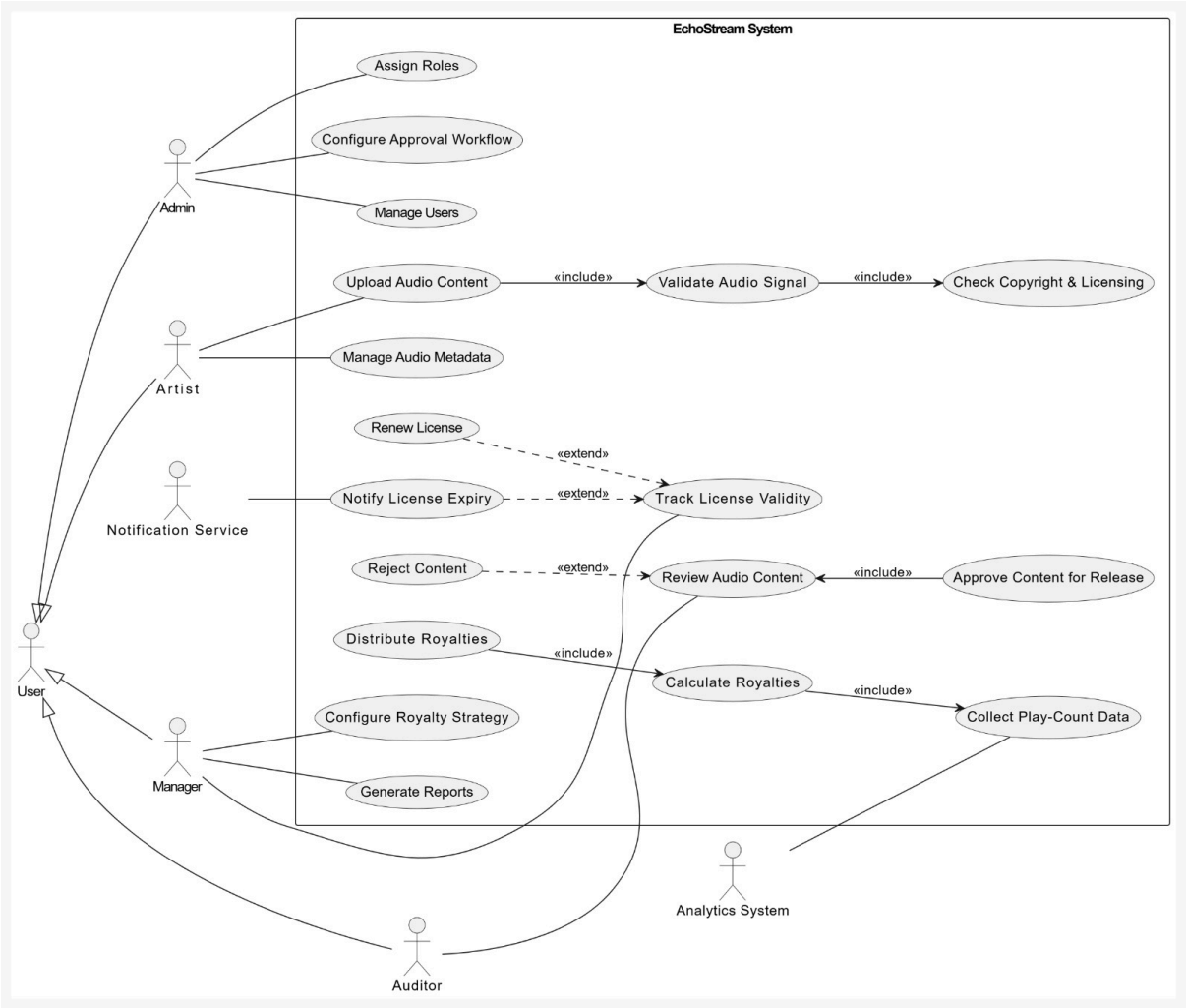
OOAD MINI PROJECT

UML DIAGRAMS

NAME	SRN
Sri Kalyan Ippili(Member 1)	PES1UG23CS593
Sreehas Adusumilli(Member 2)	PES1UG23CS591
Srirama V Swamy(Member 3)	PES1UG23CS598
Shubhang S(Member 4)	PES1UG23CS573

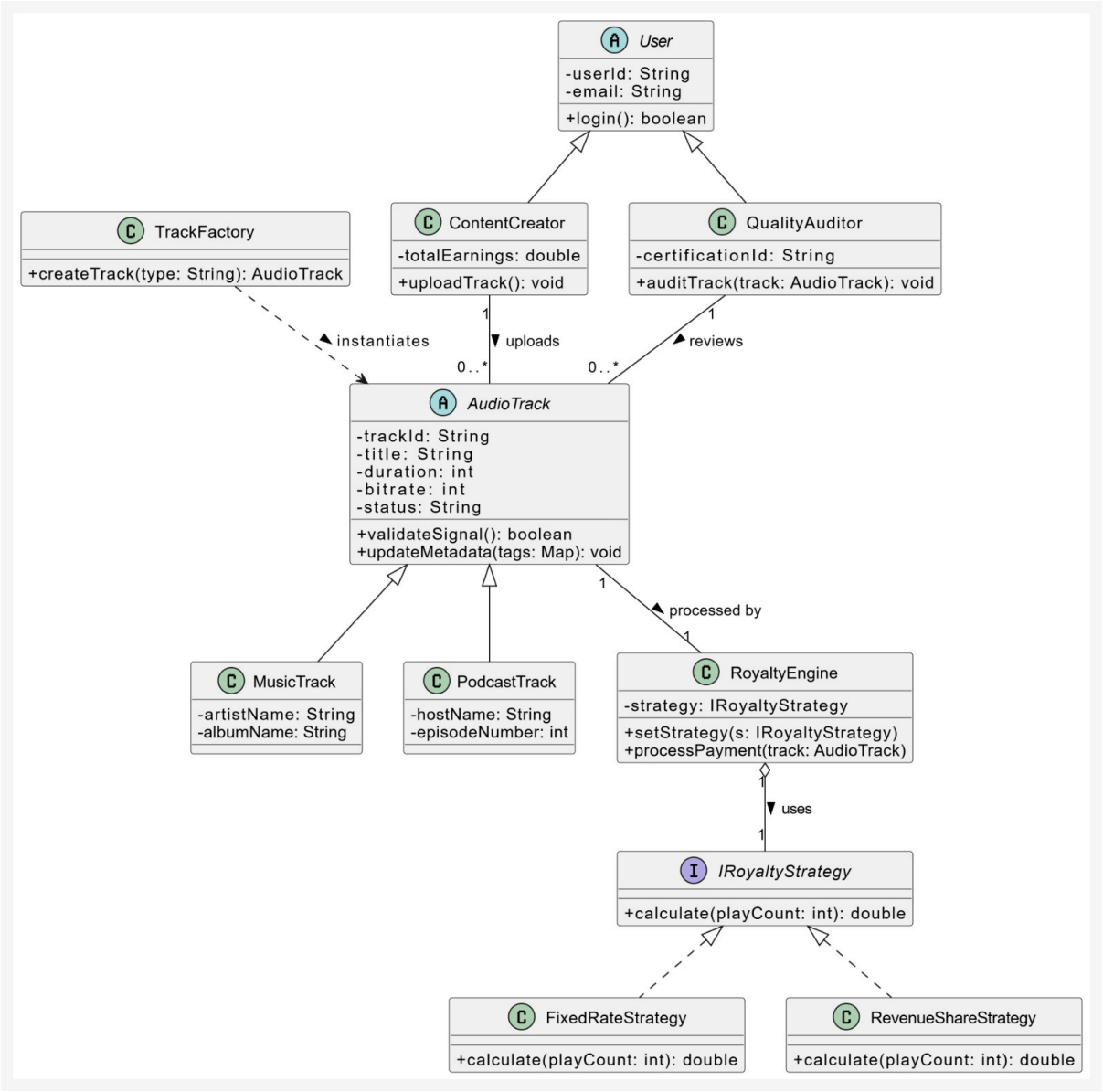
Use Case Model:

This diagram maps the complete operational workflow of the EchoStream platform, illustrating how distinct user roles—Artists, Auditors, Managers, and Admins—interact with their dedicated system verticals. It also highlights critical automated dependencies, such as how initial audio uploads mandate immediate signal and copyright validations, and how royalty distribution relies on integrated play-count analytics.

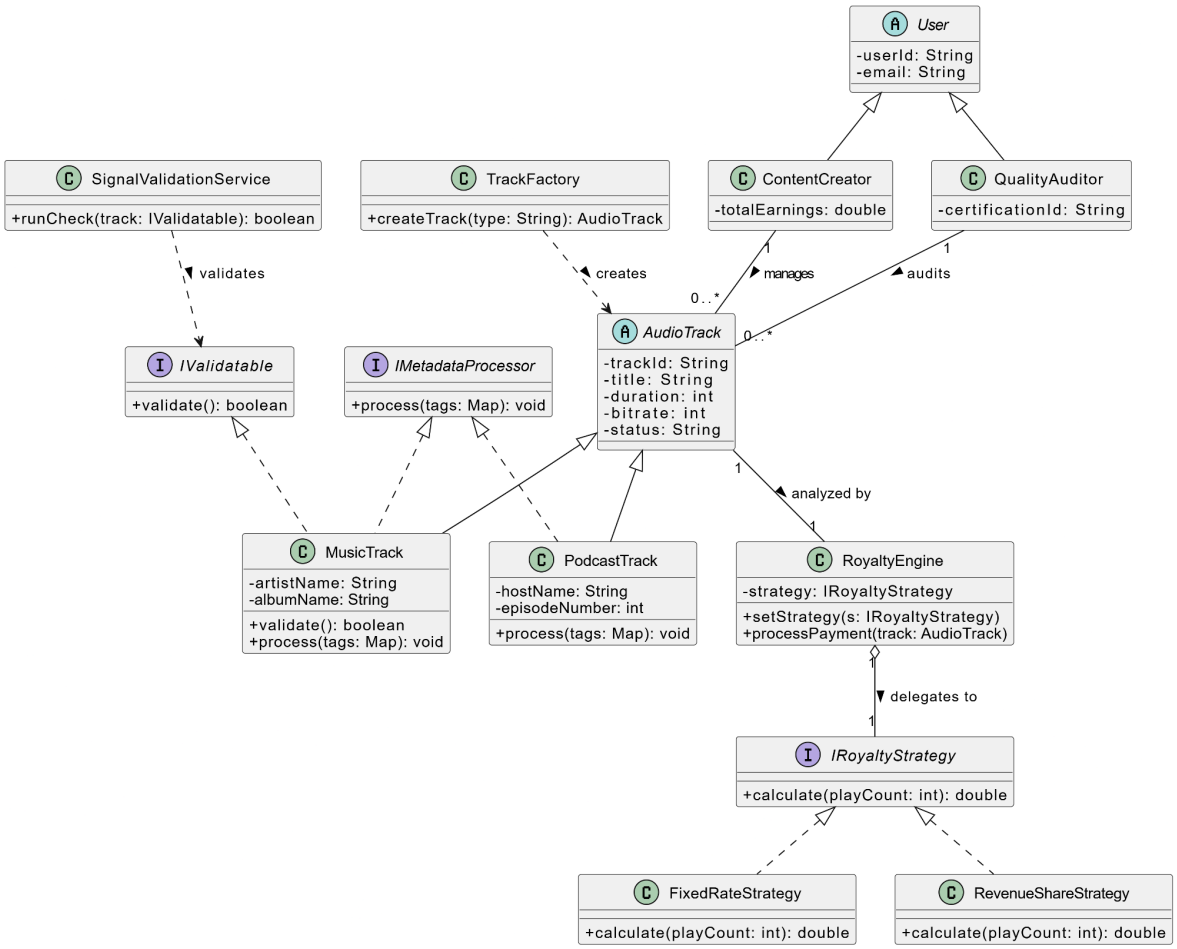


Class Diagram:

This class diagram outlines the core structural models of the EchoStream platform, defining the inheritance hierarchies and relationships between specific user roles (Creators, Auditors) and the varied audio assets they manage. Furthermore, it explicitly maps the integration of required design patterns, showcasing the Factory Method for dynamic track instantiation and the Strategy Pattern for flexible, interchangeable royalty calculations.

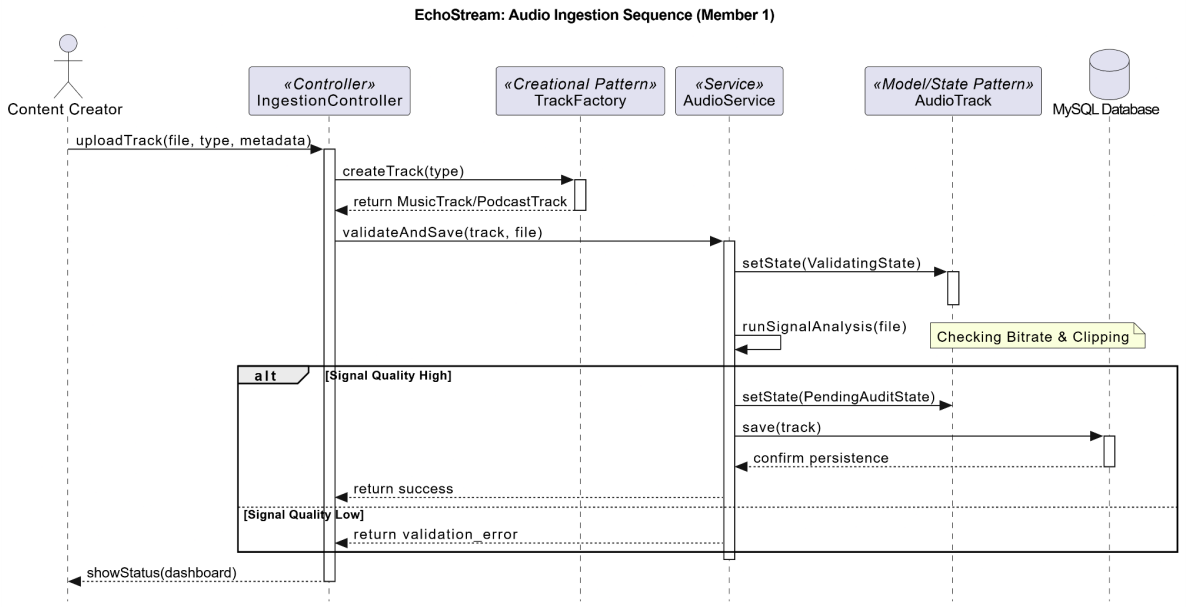


Updated Class Diagram:



Sequence Diagram:

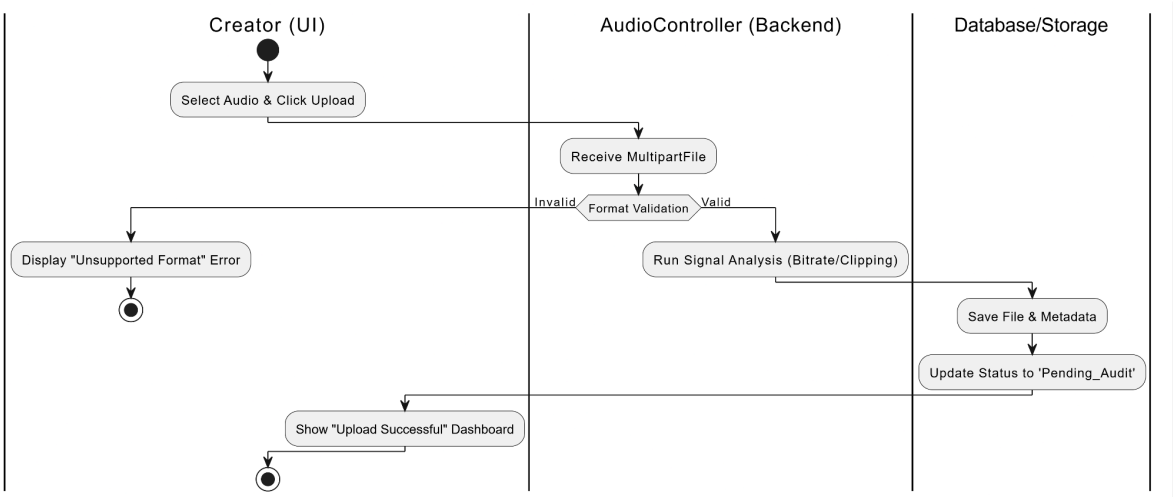
This sequence diagram illustrates the dynamic interaction between the **MVC layers** during the audio upload process, specifically showing how the **Controller** delegates object creation to the **TrackFactory**. It traces the logic from the initial user request through the automated signal validation service, ending with the **State Pattern** updating the track's status in the database.



Activity 1: Audio Ingestion (Member 1)

Focus: The flow between the Creator, the Spring Controller, and the File System.

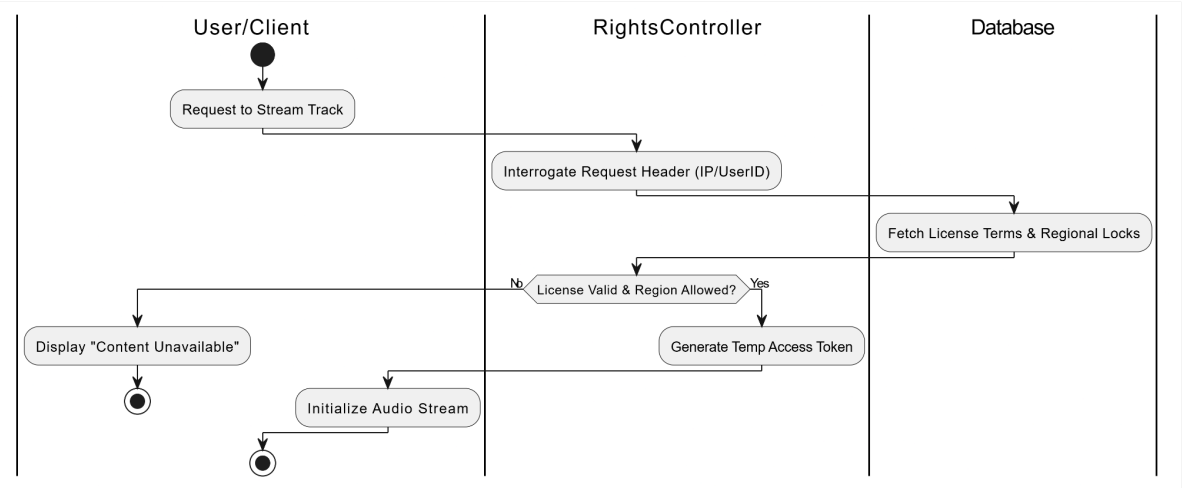
This diagram illustrates the upload workflow, where the system receives an audio file and automatically checks its format and signal quality. It highlights the clear separation between the user interface, the backend controller logic, and the database storage.



Activity 2: Rights & License Management (Member 2)

Focus: The automated verification of digital rights before granting access.

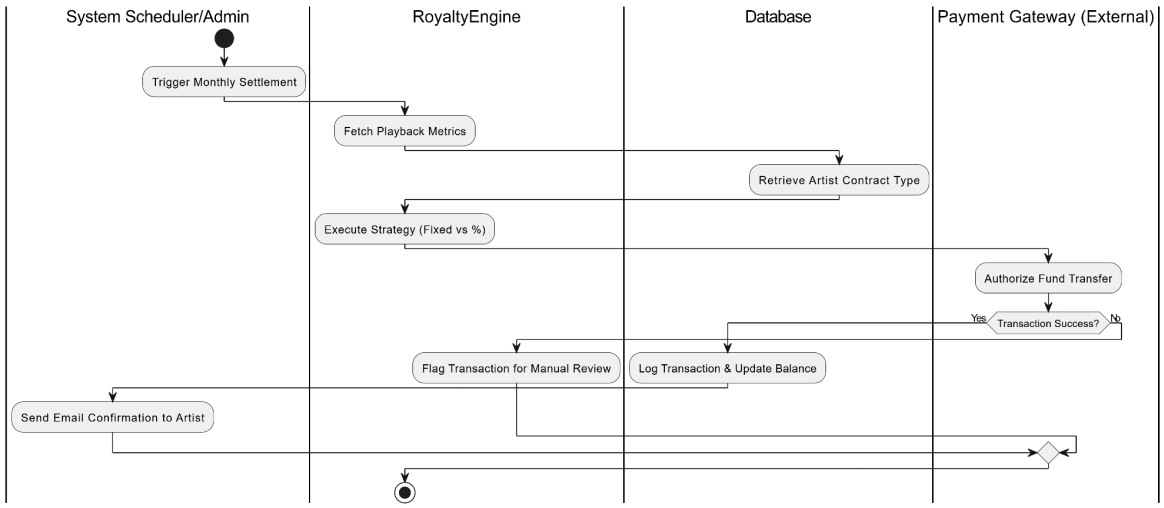
Here, the system flow for verifying digital rights is mapped out, specifically checking a user's location against the track's license terms. If the license is valid and the region is allowed, the user is successfully granted a temporary token to stream the audio.



Activity 3: Royalty Calculation (Member 3)

Focus: The interaction with an external Payment API using the **Strategy Pattern**.

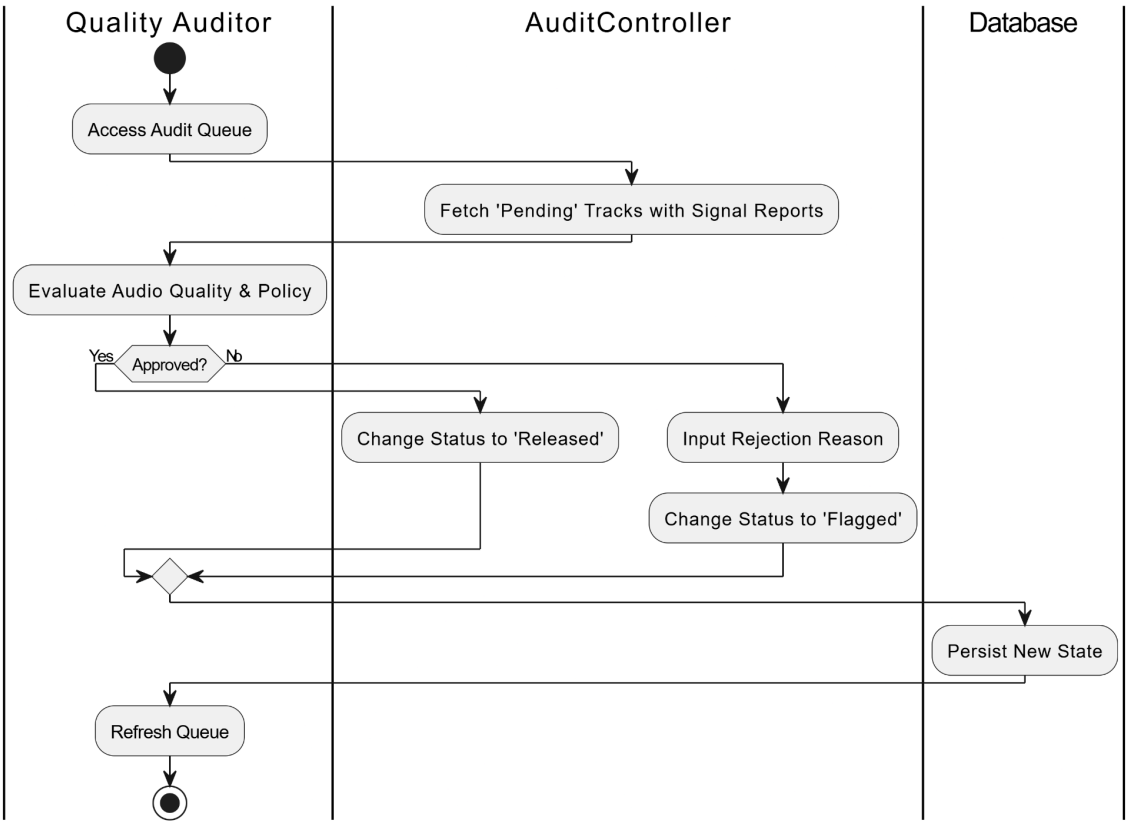
This maps out the monthly automated settlement process, utilizing a Strategy Pattern to calculate payouts based on specific artist contracts. It traces the logic from fetching playback metrics to successfully authorizing the fund transfer via an external API.



Activity 4: Audit & Approval (Member 4)

Focus: The human-in-the-loop workflow for content moderation.

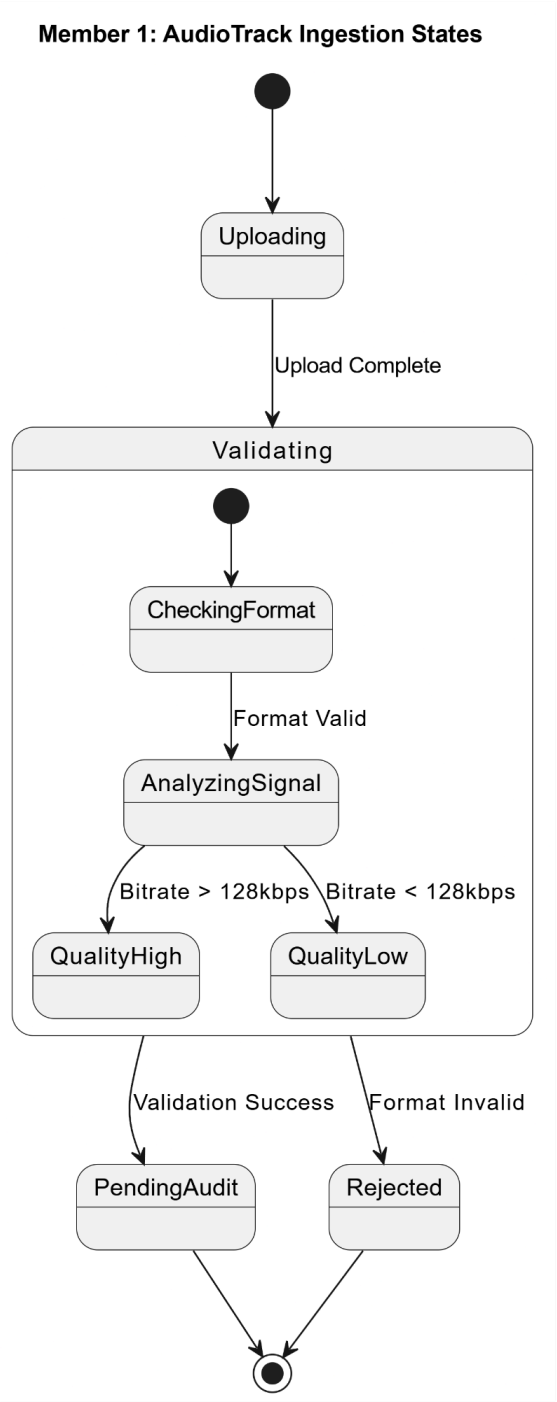
This workflow focuses on content moderation, showing how an auditor manually reviews pending tracks against quality and copyright policies. Depending on their evaluation, the track's status is updated to either 'Released' or 'Flagged' in the database.



State Diagram 1: Audio Asset Lifecycle (Member 1)

Object: `AudioTrack` This tracks the file from the initial upload through the automated quality check.

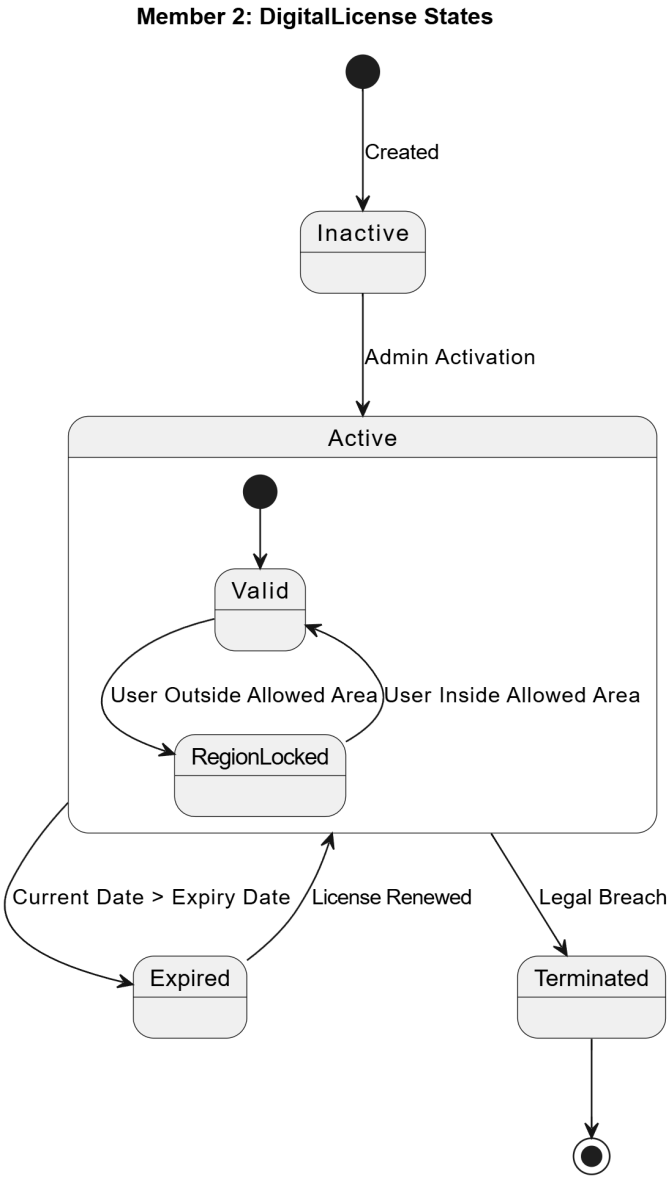
This tracks the lifecycle of an **`AudioTrack`** object during the initial ingestion phase. It details the internal validation states, such as checking formats and analyzing bitrates, before determining if the track is ready for audit or must be rejected.



State Diagram 2: License Validity (Member 2)

Object: `DigitalLicense` This manages whether a piece of content is legally "Playable" based on time and geography.

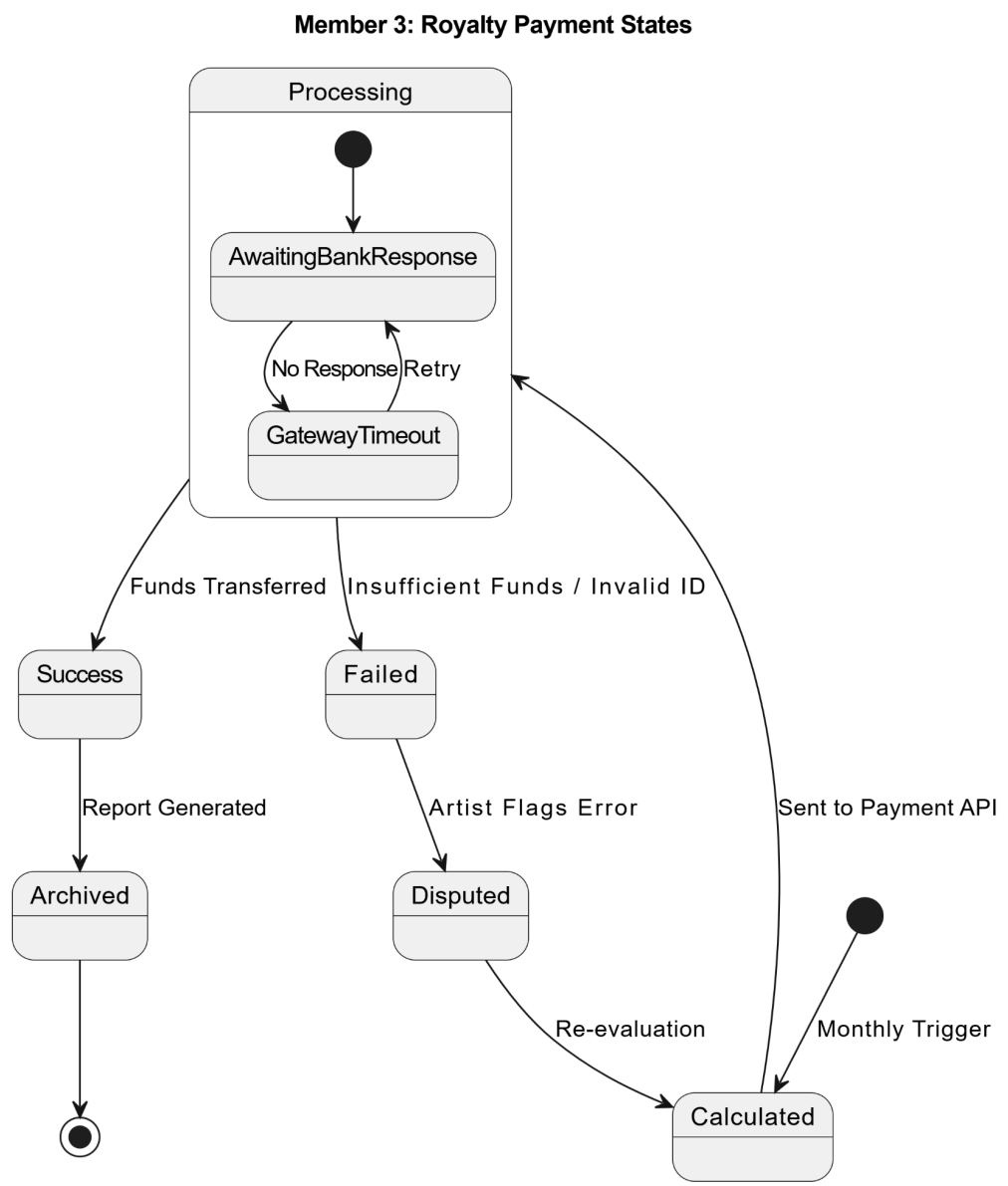
This diagram maps the **`DigitalLicense`** object to manage whether a piece of content is legally playable. It highlights the transitions between being active, dynamically region-locked, expired, or permanently terminated due to a legal breach.



State Diagram 3: Royalty Transaction (Member 3)

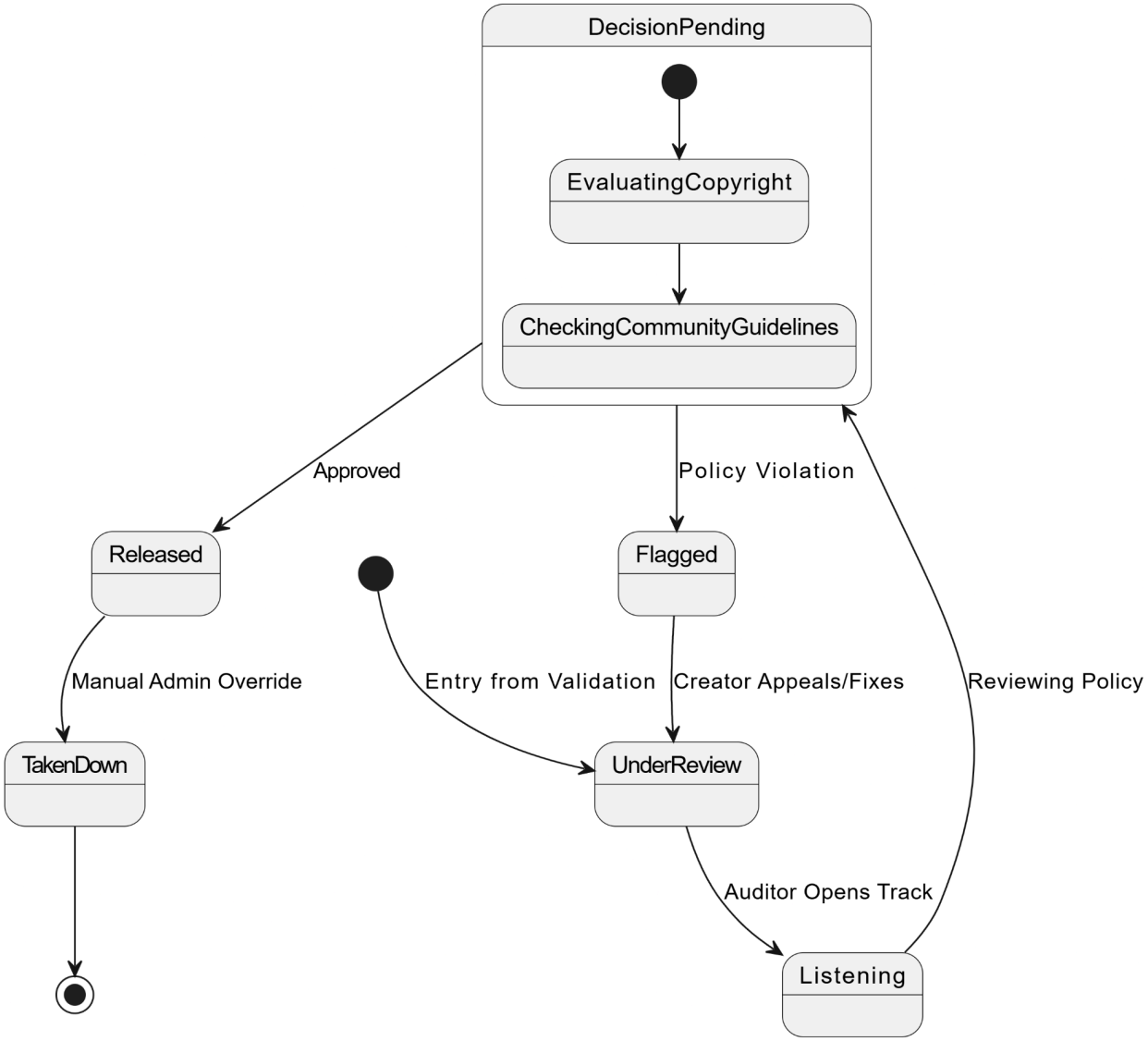
Object: `PaymentTransaction` This follows the money from calculation to the artist's bank account.

This follows a **`PaymentTransaction`** object straight through the financial processing system. It shows the specific states of awaiting bank responses, handling gateway timeouts, and resolving successful transfers versus disputed failures.



State Diagram 4: Content Approval Status (Member 4)

Object: `ApprovalWorkflow` This tracks the human auditor's decision-making process for the track's public status.

This illustrates the **`ApprovalWorkflow`** object as a human auditor determines a track's public status. It outlines the transitions from being under review and evaluated for copyright to ultimately being released, flagged, or taken down by an admin override.

Member 4: Auditor Approval States